Exercises 2.7, 2.9, 2.10

1. $\forall x \in \mathbb{R}, x^2 > 0$
   For every real number x, $x^2 > 0$
   This is false. Take $x = 0 (0 \in \mathbb{R})$, then $x^2 = 0^2 = 0$ which is not greater than 0

2. $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x^n \geq 0$
   For every real number x, there exists some n in the natural numbers such that $x^n \geq 0$
   True. Take $n = 2$, then $x^n = x^2 \geq 0$

3. $\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, ax = x$
   There exists some real number a in which, for all real numbers x, $ax = x$
   True. Take $a = 1 \Rightarrow ax = 1x = x$

9. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, m = n + 5$
   For every integer n, there is some integer m such that $m = n + 5$
   True. Taking $m = n + 5$ proves this is valid for every $n \in \mathbb{Z}$

10. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m = n + 5$
    There is some integer m such that, for every integer n, $m = n + 5$
    False. If $m = n + 5$, then for $n + 1 \in \mathbb{Z}, m = n + 5 \neq (n + 1) + 5$
    An example could be: $n = 2 \Rightarrow m = 7$, but for $n = 3, n + 5 = 8 \neq 7$

1. If f is a polynomial and its degree is greater than 2, then f' is not constant.
   $P : f$ is a polynomial
   $Q : deg(f) > 2$
   $R : f'$ is not constant

   $P \wedge Q \Rightarrow R$

2. The number x is positive but the number y is not positive.
   $P :$ The number x is positive
   $Q :$ The number y is not positive.

   $P \wedge Q$

3. If x is prime, then $\sqrt{x}$ is not a rational number.
   $P : x$ is prime
   $Q : \sqrt{x} \notin \mathbb{Q}$

   $P \Rightarrow Q$

4. For every prime number p there is another prime number q with $q > p$.

   P is the set of prime numbers, $\forall p \in P, \exists q \in P, q > p$

5. For every positive number ϵ , there is a positive number δ for which $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$

   $\forall \epsilon \in \mathbb{R}, \epsilon > 0, \exists \delta \in \mathbb{R}, \delta > 0, (|x - a| < \delta) \Rightarrow (|f(x) - f(a)| < \epsilon)$

6. For every positive number ϵ there is a positive number M for which $|f(x) - b| < \epsilon$, whenever $x > M$

   $\forall \epsilon \in \mathbb{R}, \epsilon > 0, \exists M \in \mathbb{R}, M > 0, (x > M) \Rightarrow (|f(x) - b| < \epsilon)$

7. There exists a real number a for which $a + x = x$ for every real number x.

   $\forall x \in \mathbb{R}, \exists a \in \mathbb{R}, a + x = x$

9. If x is a rational number and $x \neq 0$, then $\tan(x)$ is not a rational number.
   $P : x \in \mathbb{Q}$
   $Q : x \neq 0$
   $R : \tan(x) \in \mathbb{Q}$

   $P \wedge Q \Rightarrow (\sim Q)$

10. If $\sin(x) < 0$, then it is not the case that $0 \leq x \leq \pi$
    $P : \sin(x) < 0$
    $Q : 0 \leq x \leq \pi$

    $P \Rightarrow \sim Q$

11. There is a Providence that protects idiots, drunkards, children and the United States of America. (Otto von Bismarck)
    S is the set $\{idiots, drunkards, children, USA\}$
    $P(x) : x$ is a providence
    $Q(x, y) : x$ protects y

    $\forall y \in S, \exists x, P(x) \wedge Q(x, y)$

12. You can fool some of the people all of the time, and you can fool all of the people some of the time, but you can't fool all of the people all of the time. (Abraham Lincoln)
T is the set of the all times, P the set of all the people
$Q(x, y) =$ Fooling x people for time y

$((\exists x \in P, \forall y \in T, Q(x, y)) \vee (\forall x \in P, \exists y \in T, Q(x, y)) \wedge \sim (\forall x \in P, \forall y \in T, P(x, y))$
$= (\exists x \in P, \forall y \in T, Q(x, y)) \vee (\forall x \in P, \exists y \in T, Q(x, y)) \wedge (\exists x \in P, \exists y \in P, \sim Q(x, y))$

13. Everything is funny as long as it is happening to somebody else. (Will Rogers)
X is the set of everything
$P : x$ is funny
$Q : x$ is happening to you
$R : x$ is happening to someone

$\forall x \in X, (\sim Q \wedge R \Rightarrow P)$

1. The number x is positive, but the number y is not positive.

   $\sim (x > 0 \wedge y \leq 0) = \sim (x > 0) \vee \sim (y \leq 0) = (x \leq 0) \vee (y > 0)$
   x is not positive or y is positive

2. If x is prime, then $\sqrt{x}$ is not a rational number.

   $\sim ((x \text{ is prime}) \Rightarrow (\sqrt{x} \notin \mathbb{Q})) = (x \text{ is prime}) \wedge \sim (\sqrt{x} \notin \mathbb{Q}) = (x \text{ is prime}) \wedge (\sqrt{x} \in \mathbb{Q})$
   There exists some prime x where $\sqrt{x}$ is a rational number

3. For every prime number p, there is another prime number q with $q > p$.

   Define $P[\mathbb{N}]$ as the set of all primes
   $\sim (\forall p \in P, \exists q \in P, q > p) = \exists p \in P, \forall q \in P, q \leq p$
   There exists some prime p such that for all primes q, $q \leq p$

4. For every positive number ϵ, there is a positive number δ for which $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$

   $\sim (\forall \epsilon \in (0, \infty), \exists \delta \in (0, \infty), (|x - a| < \delta) \Rightarrow (|f(x) - f(a)| < \epsilon))$
   $\exists \epsilon \in (0, \infty), \forall \delta \in (0, \infty), \sim ((|x - a| < \delta) \Rightarrow (|f(x) - f(a)| < \epsilon))$
   $\exists \epsilon \in (0, \infty), \forall \delta \in (0, \infty), (|x - a| < \delta) \wedge \sim (|f(x) - f(a)| < \epsilon)$
   $\exists \epsilon \in (0, \infty), \forall \delta \in (0, \infty), (|x - a| < \delta) \wedge (|f(x) - f(a)| \geq \epsilon)$
   There exists some $\epsilon > 0$ such that for every $\delta > 0$, there is a number x where $|x - a| < \delta$ and $|f(x) - f(a)| \geq \epsilon$

5. For every positive number ϵ there is a positive number M for which $|f(x) - b| < \epsilon$, whenever $x > M$

   $\sim (\forall \epsilon \in (0, \infty), \exists M \in (0, \infty), ((x > M) \Rightarrow (|f(x) - b| < \epsilon)))$
   $\exists \epsilon \in (0, \infty), \forall M \in (0, \infty), (x > M) \wedge (|f(x) - b| \geq \epsilon)$
   There exists some positive ϵ such that for all positive M, there is a number $x > M$ where $|f(x) - b| \geq \epsilon$

6. There exists a real number a for which $a + x = x$ for every real number x.

   $\sim (\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, a + x = x)$
   $\forall a \in \mathbb{R}, \exists x \in \mathbb{R}, a + x \neq x$
   For every real number a, there exists a real number x such that $a + x \neq x$

9. If $\sin(x) < 0$,then it is not the case that $0 \leq x \leq \pi$.

   $\sim(\sin(x) < 0 \Rightarrow \sim (0 \leq x \leq \pi))$
   $= (\sin(x) < 0) \wedge (0 \leq x \leq \pi)$
   There exists some value $x \in [0, \pi]$ such that $\sin(x)$ is negative

10. If f is a polynomial and its degree is greater than 2, then f' is not constant.

    $\sim (((f \text{ is a polynomial}) \wedge (deg(f) > 2)) \Rightarrow (f' \text{ is not constant}))$
    $= (f \text{ is a polynomial}) \wedge (deg(f) > 2) \wedge (f' \text{ is a constant})$
    There exists a polynomial f such that the degree of f is greater than 2 and f' is a constant